

Genetics I

THE FIELD OF BIOLOGY THAT INVESTIGATES INHERITANCE OF CHARACTERISTICS FROM PARENTS IS CALLED GENETICS.

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◀ 80–81 Evolution

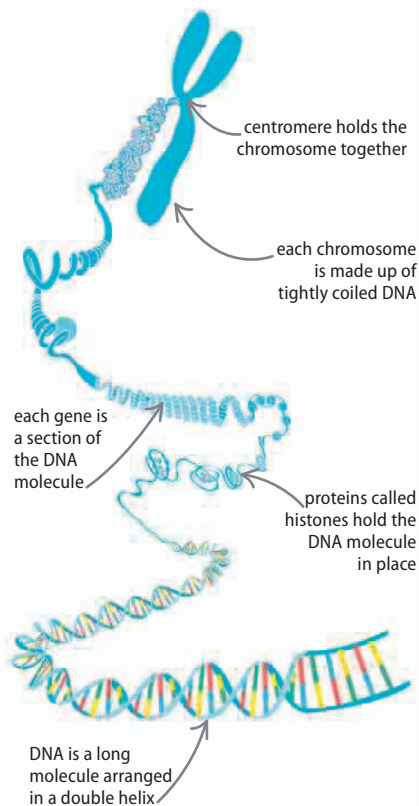
Polymers

162–163 ▶

The instructions for making a living body are called genes. Each gene relates to a specific characteristic, such as eye color or height. A full set of genes is inherited from both parents, so a child shares many of his or her parents' characteristics.

Chromosomes

Genes are carried on long chemical chains of deoxyribonucleic acid (DNA). DNA is stored inside a cell's nucleus on chromosomes, which are the vehicles that carry the genes as they pass from one generation to the next. The number of chromosomes in a cell is called the diploid number. Sperm and eggs contain a half-set, or haploid number, of chromosomes.



Genes and alleles

Each gene has a specific position on its chromosome. Everyone has two versions of each chromosome, one from each parent. That means they have two versions, known as alleles, of each gene. The two alleles form a person's genotype. One allele is often dominant over the other, which is recessive, so just one characteristic (known as a person's phenotype) is expressed.

▽ Genetic probability

In this example, we see two parents and their possible offspring. Both parents give one allele to their children. The allele for brown eyes (B) is dominant, and the allele for blue eyes (b) is recessive. The mother has a recessive allele, so it is equally likely that they have a brown or a blue-eyed child.

